Design review #2 – Ring oscillator

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1 General information

The primary aim of this project is to develop a high accuracy, low noise, space graded INA. In parallel with that, a 6b tunable ring oscillator (RO) has been developed to provide the following support:

- Simple verification of the I2C / SPI by sweeping codes and observing the change in oscillation frequency (f_{osc}).
- Simple correlation between silicon and simulation (i.e. iHP device models) by comparing f_{osc} over all codes pre and post silicon.
- Physical verification flow (LVS / DRC / PEX) for this project was pipe cleaned and verified using this simple (quick to simulate) block.
- Transition from iHP's SG13G2 → SG13CMOS5L process models was pipe cleaned and verified using this simple (quick to simulate) block.
- Since the RO output is brought off chip, it provides a nice demonstration of the project's capabilities and interaction of the analog and digital blocks together on chip – user can see an oscillating waveform on an oscilloscope whose f_{osc} varies with the application of a trim code.
- An RO can be used as a crude temperature sensor. As accurate temperature sensors will be placed on the ASIC, the performance of an "RO based temperature sensor" can be accurately characterized to lay the foundations for a future possible project.

1.1 Project goal

To show the capability of open source tools to develop industry quality ASICs, verified across industry standard conditions and taped out on silicon from a European fab (IHP).

1.2 Specifications

Below specifications to be met across full process (SS/FF/SF/FS), voltage (±5%) and temperature (-40 → 125C) (i.e. PVT) conditions.

Parameter	Unit	Value	Comment
vdda_lv	V	1.5	Digital supply
vssa	V	0	Ground
idda_hv	mA	1	Digital supply current
Power	mW	1.5	Digital power consumption
Fosc (typical conditions)	MHz	10	TT/27C/mid-code
Resolution	%	1	Accuracy
Fosc spread	± %	5	
Duty cycle max	%	55	
Duty cycle min	%	45	
Dfosc	± %	20	Tunability

Table 1: RO specifications

1.3 Purpose of Design review #2 (DR2)

Purpose of DR2 is to review final schematics before delivery to layout. Therefore, upon approval of DR2, schematics are “frozen” and layout commences.

1.4 Repository details

All material relating to this project can be found in the below publicly available repository:

https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main

Screenshot of the project landing page is shown in Figure 1.

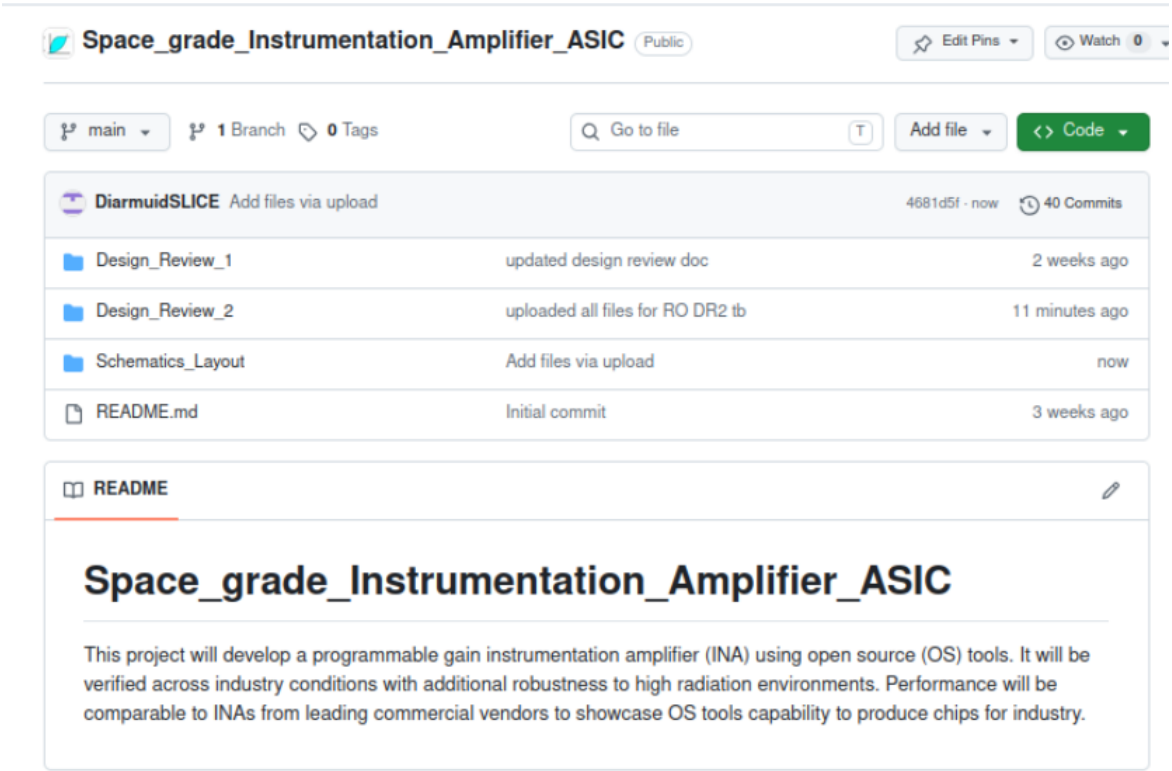

Figure 1 Project landing page

As this document pertains to DR2, all material relating to it is found in the sub directory “Design_Review_2”. As block schematics relate to all DR’s, these are contained in the separate folder “Schematics_Layout”. As shown in Figure 2 this folder contains a sub-directory to each block in the ASIC design.

Inside each folder here, lives the following views:

- Schematic (to be frozen at DR2)
- Symbol (to be frozen at DR2)
- Layout (to be frozen at DR3)

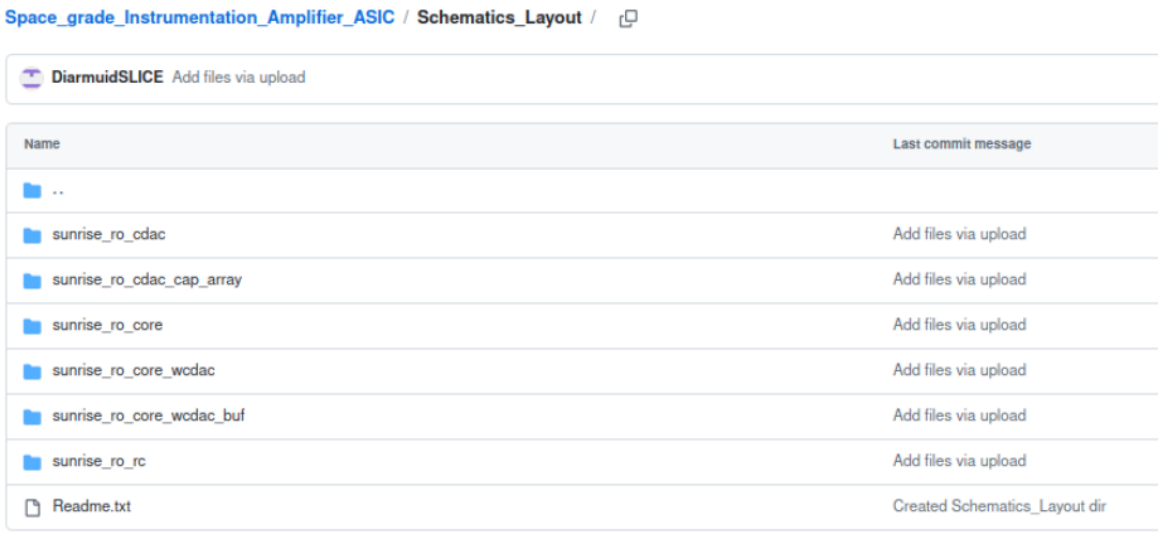

Figure 2 Project design blocks

1.5 Open source tools

The open source tools used for DR2 are summarized in Table 2, with links to install each tool provided as references.

Tool	Description
ngspice	Spice simulator [1]
xschem	Schematic / waveform viewer [2]
openVAF	verilogA compiler [3]

Table 2: Open source tools used for DR1

1.6 Open source PDK

The open source process development kit (PDK) used for the design is from IHP, process option – SG13CMOS5L [4]. This is a derivative of their more general option – SG13G2 [5] as it excludes the following:

- Top metal layer 2
- Linear caps (MiM or MoM)
- Deep Nwell (DNW)
- HBT devices

The first 3 items impacted the design but not the 4th since the design is not RF. SG13CMOS5L option was chosen primarily because its tapeout date of 13/07/26 suited the project timeline. In addition, it is lower cost (max = €1.5k/mm2 as opposed to €2.8k/mm2 for SG13G2) and runs though the fab quicker (5 months instead of 6) to achieve a fab out date in 16/12/26 [6].

2 Top-Level design description

2.1 Architecture

Any ring oscillator consists of 'N' inverting stages where N ≥ 3, as shown in Figure 3.

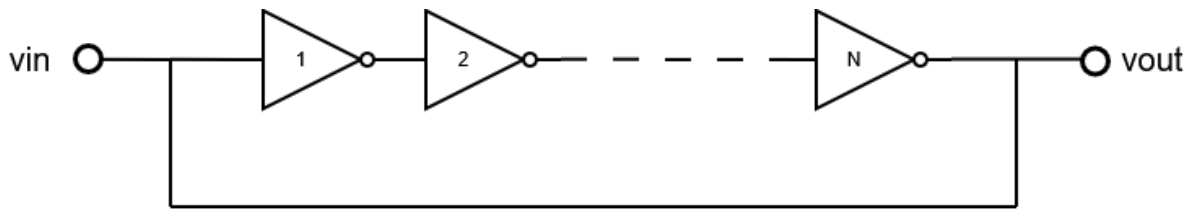

Figure 3 Basic ring oscillator architecture

The resulting oscillation frequency is defined as:

$$f_{osc} = \frac{1}{2t_{pd}N} \dots (1)$$

where:

- t_{pd} = propagation delay through an inversion stage

As t_{pd} is the time it takes the output of an inversion stage to reach vdd/2, it can also be expressed as:

$$t_{pd} = 0.693\tau = 0.693R_{op}C_{in} \dots (2)$$

where:

- R_{op} = effective output resistance of the inverter
- C_{in} = input capacitance of the inverter

Note: in (2), R_{op} varies as a function of the input signal swing due to changes in v_{gs}

Using typical values of $C_{in} = 20fF$, $R_{op} = 5k\Omega$, for N = 5 => $f_{osc} \approx 3GHz$. To get down to the 10MHz spec would require N ≈ 1000 which is an impractically large number that would result in enormous power consumption. A more efficient approach to reducing the frequency of a ring oscillator is to insert an RC network at each stage, as shown in Figure 4.

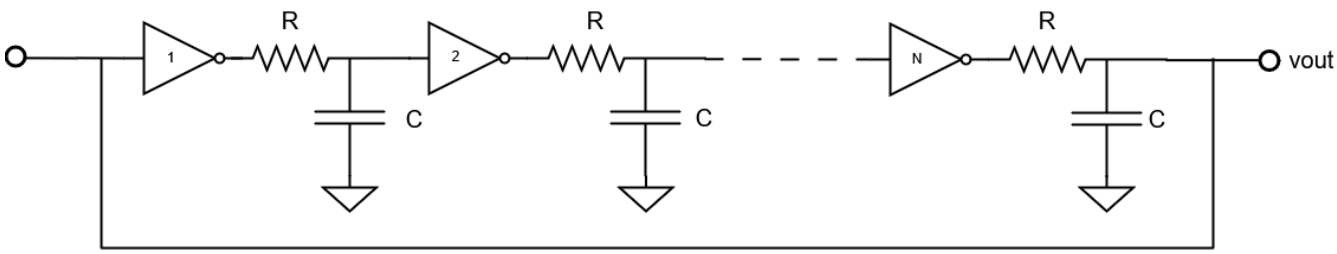

Figure 4 RC based ring oscillator

By sizing $R \gg R_{op}$ and $C \gg C_{in}$, using (1) and (2), f_{osc} can be approximated as:

$$f_{osc} = \frac{1}{2 * 0.693 * RC * N} \dots (3)$$

RC can then simply be sized accordingly for a given frequency.

Note: An interesting observation is that Δf_{osc} in Figure 4 is relatively independent on vdd while Δf_{osc} in Figure 3 varies massively with it (MHz/mV sensitivity). This gives the architecture of Figure 4 far superior power supply rejection (PSR) than that of Figure 3.

The output frequency of any ring oscillator (RO) will vary across PVT conditions where this spread in f_{osc} (Δf_{osc}) is fundamentally different between Figure 3 and Figure 4.

In Figure 3, Δf_{osc} is dominated by ΔR_{op} and so becomes a function of Δv_{th}, vdd, and temperature (T). The minimum f_{osc} (f_{min}) thus occurs at SS, LT, vdd_{min} corners with f_{max} occurring at FF, HT, vdd_{max} corners, where typical spreads are $\approx \pm 30\%$.

In Figure 4, Δf_{osc} is purely a function of ΔR (and hence T) and ΔC. f_{min} thus occurs at R_{max}, C_{max} corners while f_{max} thus occurs at R_{min}, C_{min} corners, where typical spreads are $\approx \pm 20\%$.

To account for these spreads and achieve the target f_{osc} across PVT conditions, a frequency tuning mechanism (or frequency trim) must be included in the architecture whose trim range spans the expected Δf_{osc} range.

Two methods exist for tuning a ring oscillator (RO). The first is by current starving whereby the current into each inverter stage is adjusted to give a linear change in frequency. The second is by placing a CDAC on the RO output as shown in Figure 5.

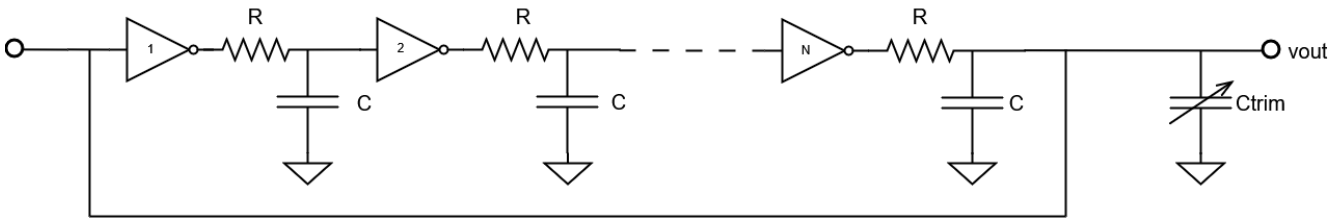

Figure 5 Trimmable RC based ring oscillator

With Ctrim present, f_{osc} is described as:

$$f_{osc} = \frac{1}{C_{trim} + (2 * 0.693 * RC * N)} \ldots (4)$$

Due to its simplicity, the CDAC based approach was chosen using a 6b CDAC.

3 Testbench description

3.1 Testbench details

One testbench has been used to verify the design meets the performance specified in

Table 1. This been checked into the repository detailed in section 1.4, as shown in Figure 6.

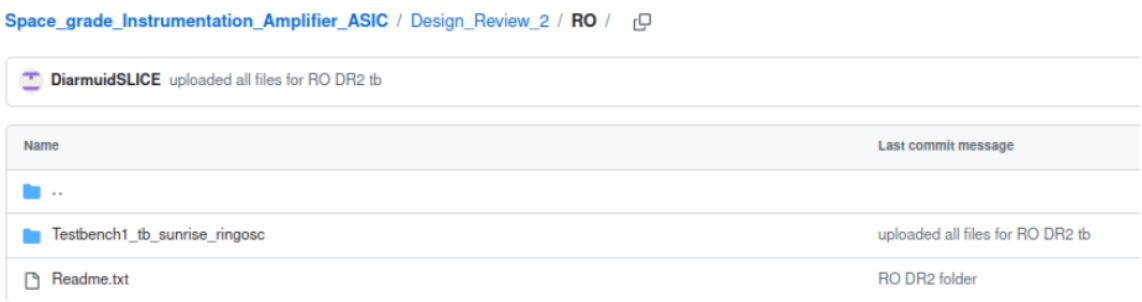

Figure 6 Testbench checked into the repository

Testbench name corresponds to the top level schematic on which to run the testbench (e.g. In folder “Testbench_1_tb_sunrise_ringosc”, run file tb_sunrise_ringosc.sch in xschem). To run the testbench, assuming the tools detailed in section 1.5 have been installed, simply download the corresponding folder, open <folder_name>.sch in xschem and hit ‘s’ (for simulate) .

Testbench schematic is shown in Figure 7 where the RO is shown in the yellow box. Trim codes to the RO are supplied by a verilogA decimal to binary encoder, shown in the green box.

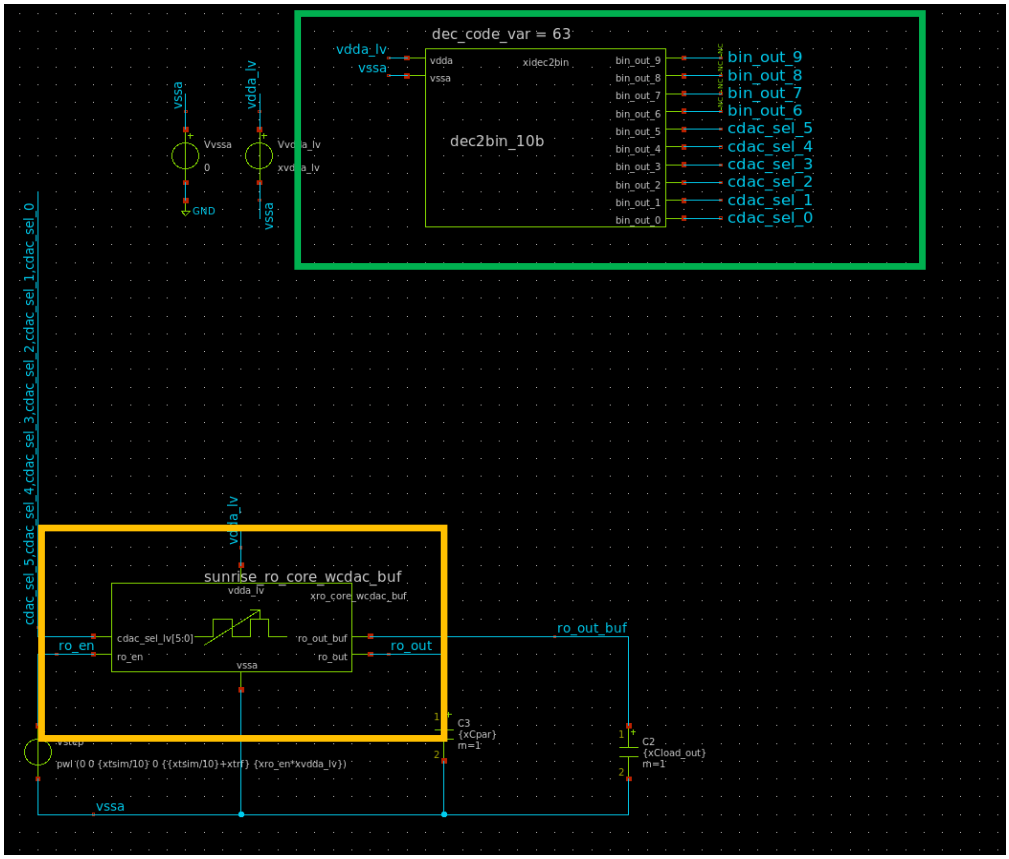

Figure 7 RO testbench schematic

4 Design hierarchy

For ease of layout and physical verification (PV) checks, design has been constructed into the multi-layer hierarchy summarized in Table 2.

Level	1	2	3	4
	sunrise_ro_core_wcdac_buf	sunrise_ro_core_wcdac	sunrise_ro_core	
				sunrise_ro_rc
				sg13cmos5l_nand2
				sg13cmos5l_inv8
				sg13cmos5l_inv16
			sunrise_ro_cdac	
				sunrise_ro_cdac_cap_array
		sg13cmos5l_inv4		
		sg13cmos5l_inv16		

Table 2 Design hierarchy

Schematics for levels 1 → 4 are shown in Figure 8 → Figure 13 respectively.

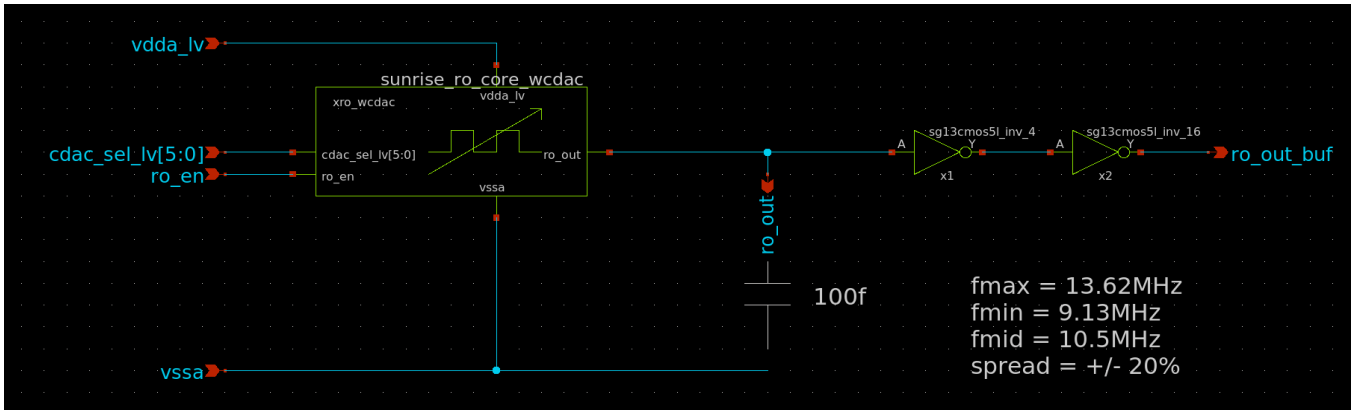

Figure 8 Level 1: sunrise_ro_core_wcdac_buf

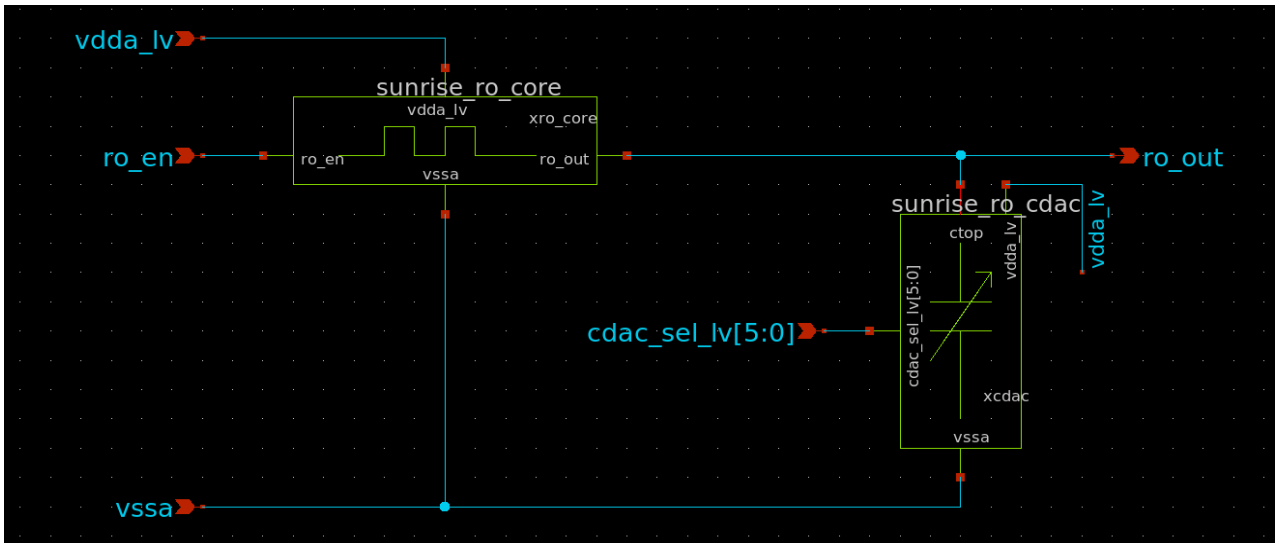

Figure 9 Level 2: sunrise_ro_core_wcdac

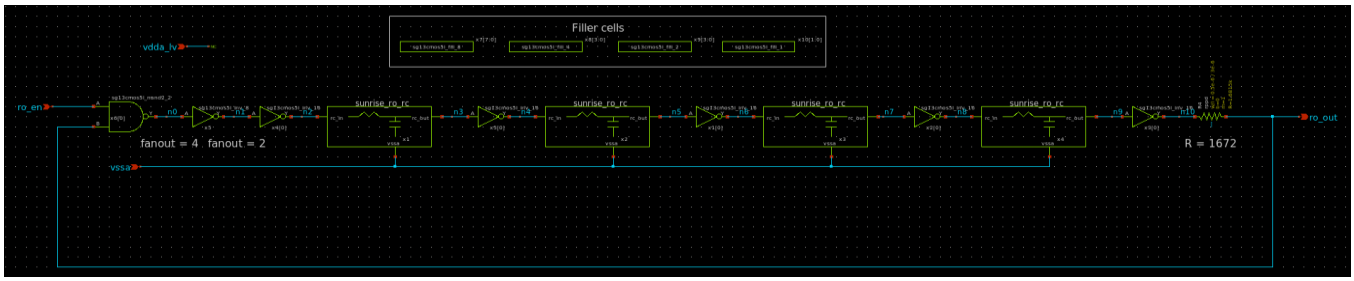

Figure 10: Level 3: sunrise_ro_core

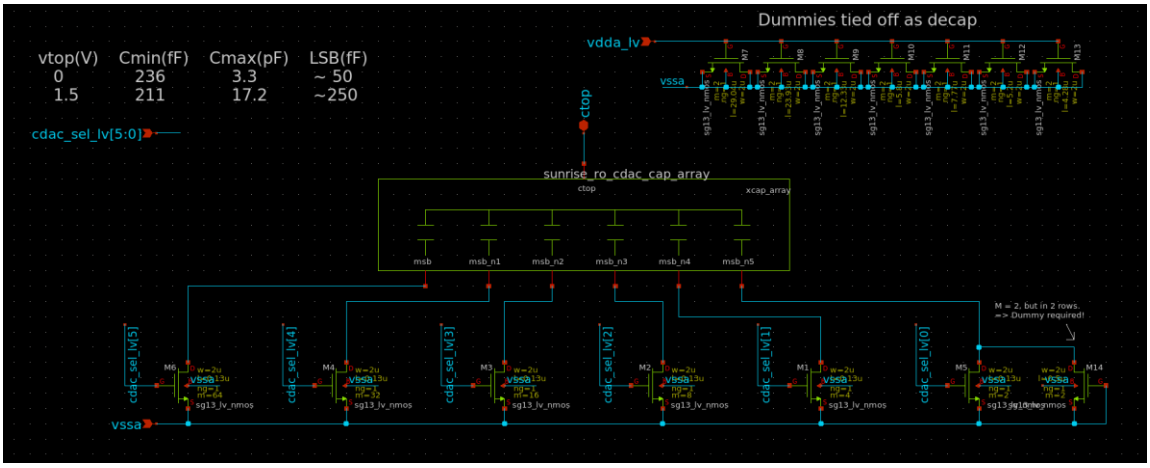

Figure 11: Level 3: sunrise_ro_cdac

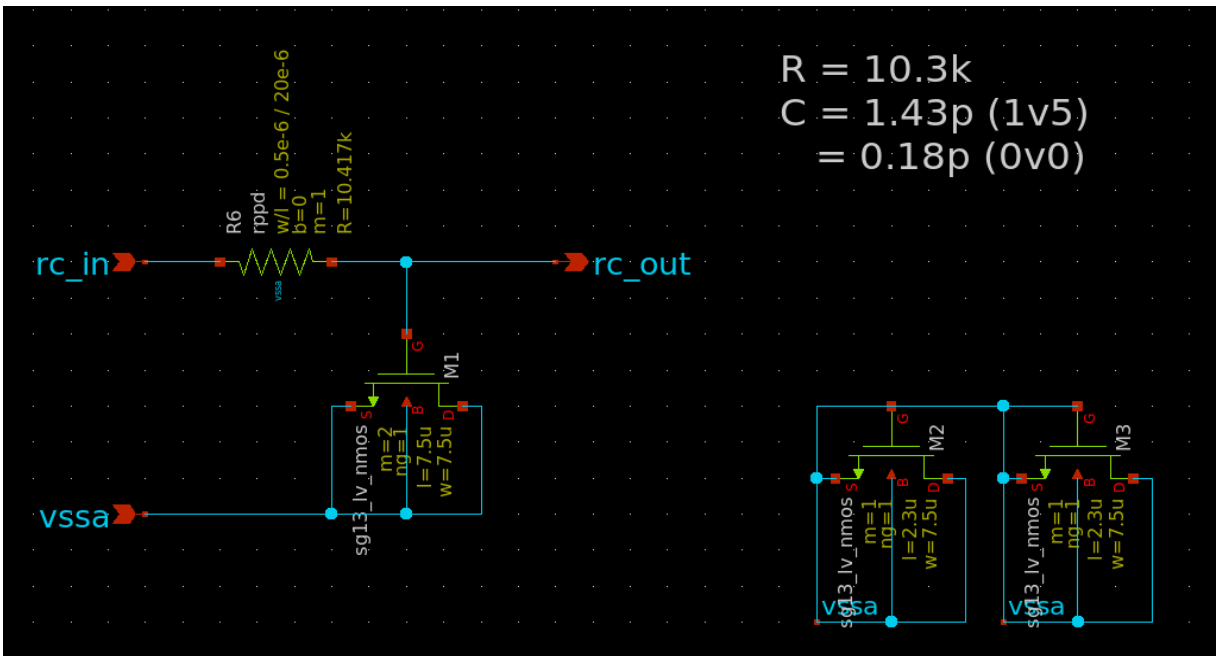

Figure 12 Level 4: sunrise_ro_rc

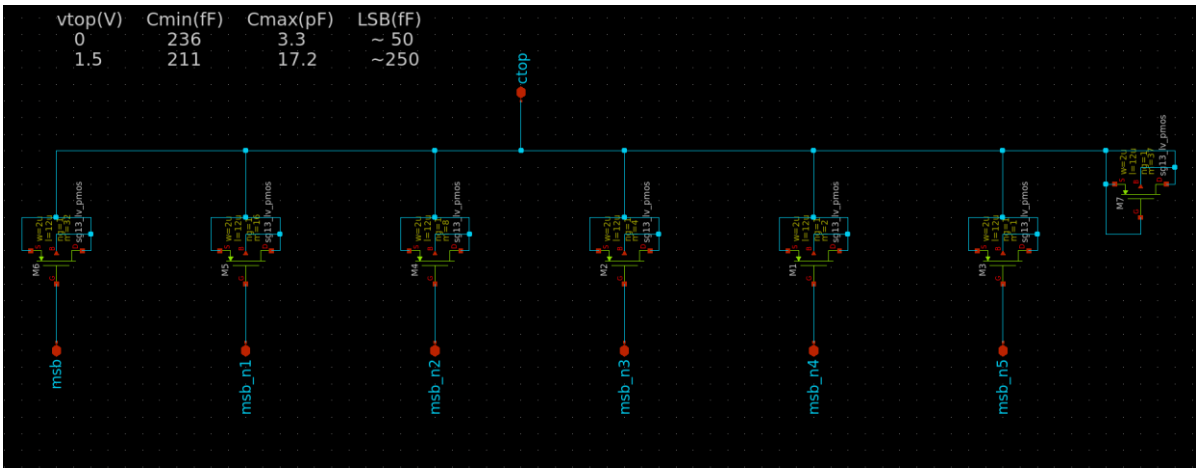

Figure 13 Level 5: sunrise_ro_cdac_cap_array

5 Testbench simulation results

5.1 Functionality

Figure 14 shows the buffered and unbuffered outputs from the block under TT/27C conditions for the mid trim code (dec' 31).

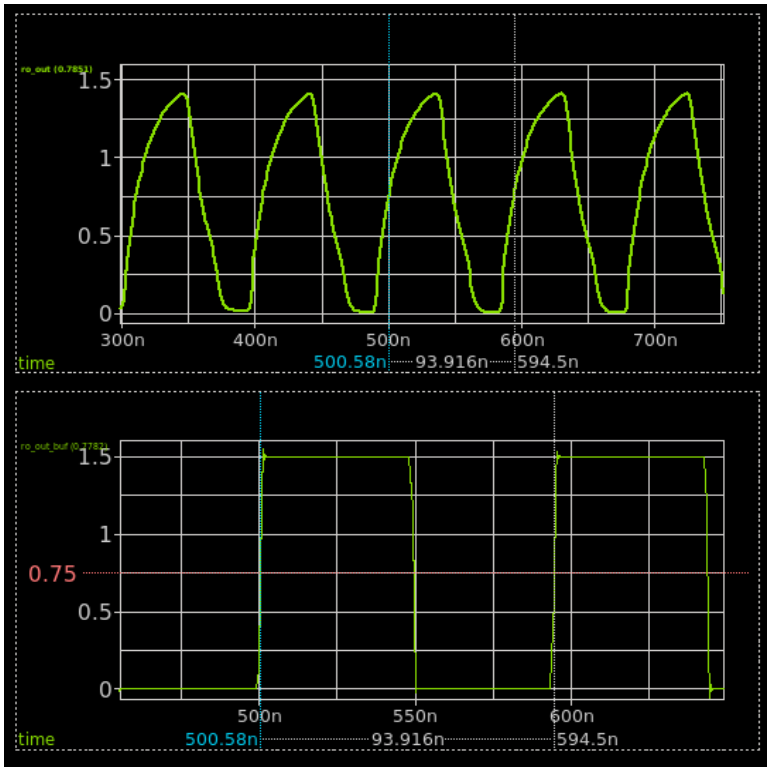

Figure 14 RO outputs

As is typical for any RO, the unbuffered output has an RC characteristic that gets converted to a 50% duty cycled square wave after it passes through a buffer. Outputs from the testbench measures are shown in Figure 15.

```
--
vdda_lv = 1.5
--
idda_uA = 556.15
freq_MHz = 10.5639
D_cyc = 52.0511
ro_out_max = 1.49986
ro_out_min = 0.00160382
ro_out_p2p = 1.49825
n0_p2p = 1.5579
n1_p2p = 1.55691
n2_p2p = 1.55611
n3_p2p = 1.49844
n4_p2p = 1.50043
n5_p2p = 1.43044
n6_p2p = 1.50041
n7_p2p = 1.49776
n8_p2p = 1.49957
n9_p2p = 1.42745
n10_p2p = 1.50015
--
```

Figure 15 Testbench measures

From Figure 15 the output frequency can be seen to be ~ 10.6MHz. It is intentionally > 10MHz to allow for a reduction post layout due to parasitics.

Finally, the peak to peak (Vp2p) swing at each stage output is shown to be ~ vdd which verifies sufficient drive strength has been maintained throughout the chain of inverters.

5.2 Frequency trim

Trimability of the RO was next analysed under TT/27C conditions by sweeping the 6b CDAC trim code from 0 → 63 and measuring f_{osc} for each trim code. Resulting tuning curve is shown in Figure 16.

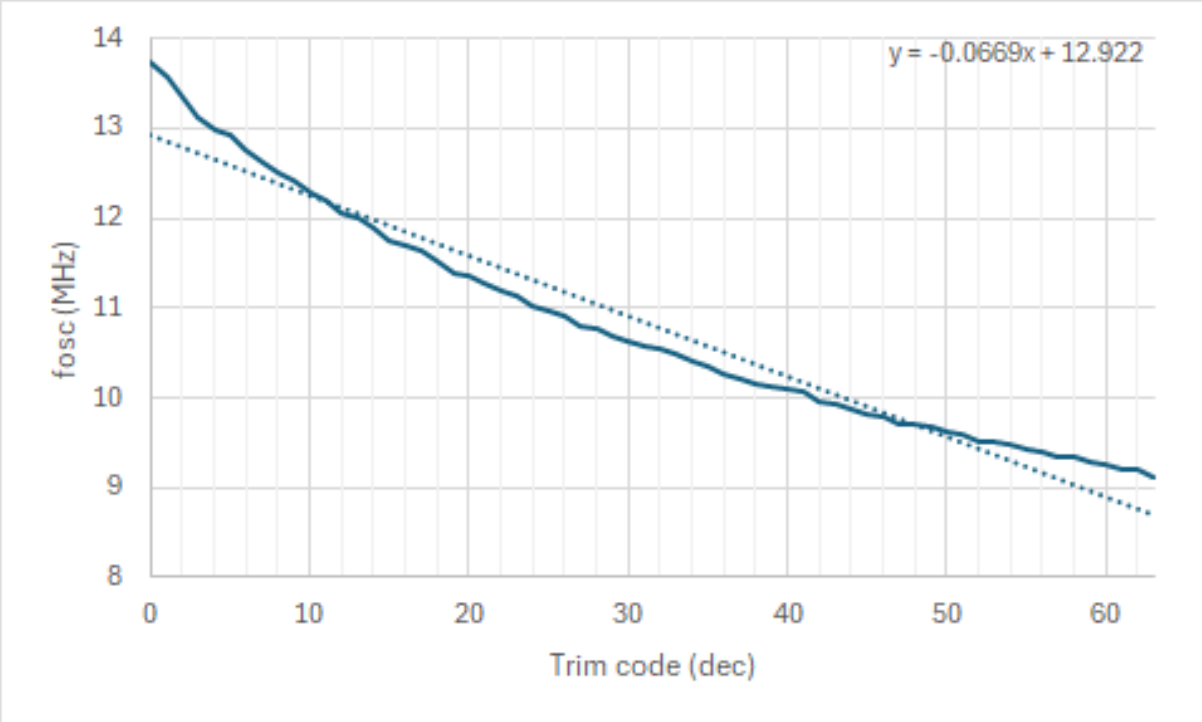

Figure 16 RO tuning curve

Figure 16 shows f_{osc} = 9.13 → 13.56MHz, representing a tuning range of ± 19.5%. Although this is ± 0.5% out of spec, as shown in section 5.3, it is large enough to cover the PVT spread of the RO.

Figure 16 also shows the tuning curve to be non-linear. This is completely expected and occurs because, as shown in (4), $f_{osc} \propto 1/C_{trim}$ thereby resulting in a 2nd order non-linearity. This is the primary disadvantage with the "brute force" method of tuning using a CDAC which doesn't occur in current starved architectures (since $f_{osc} \propto I_{trim}$). As a result, the LSB is non-linear ranging from 23 → 244kHz with an average value, as shown in Figure 16, of 67kHz.

The non-linearity can be quantified by looking at the RO's differential non-linearity (DNL) and integral non-linearity (INL) plots shown in Figure 17 and Figure 18 respectively.

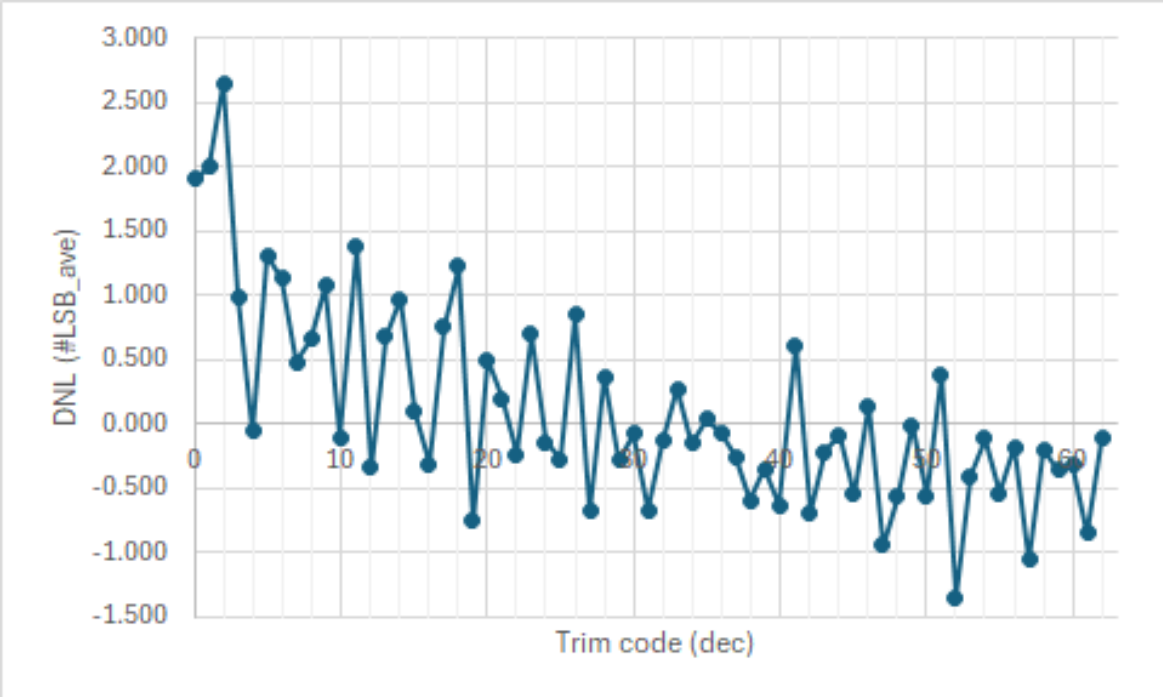

Figure 17 RO DNL

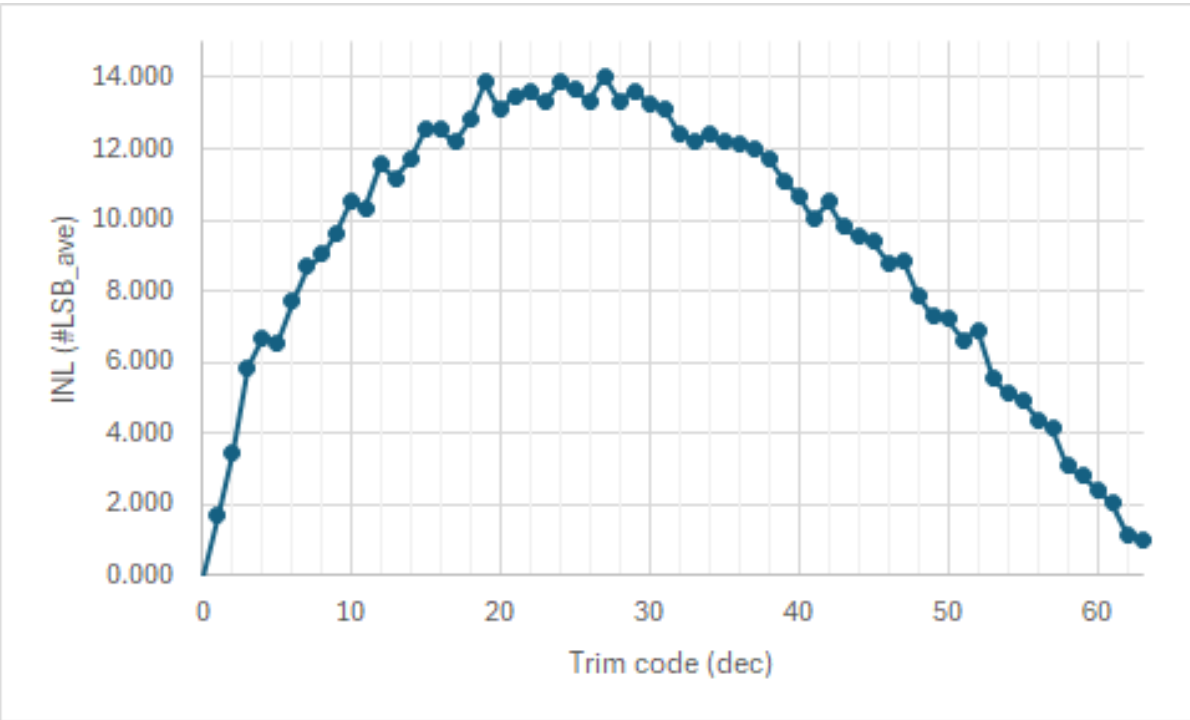

Figure 18 RO INL

From Figure 18 we can see the 2nd order non-linear characteristic more clearly and that it can produce up to 14LSB's (at 6b) of non-linearity at the mid-code. To ensure mid-code produces $f_{osc} \approx 10MHz$, this non-linearity had to be taken into account by re-centering the tuning curve around 10MHz at the mid-code.

RO resolution is found from the LSB's. Taking the average LSB of 67kHz gives a resolution of $(67k/10M) = 0.67\%$. Taking the max LSB of 244kHz then gives a resolution of 2.44%. Since taking the max LSB is overly pessimistic, the average resolution was specified at 1%, which is clearly met.

Figure 19 shows the min and max RO frequencies at mid code (dec'31).

```
vdda_lv = 1.5
--
idda_uA = 346.109
freq_MHz = 12.355
D_cyc = 52.9189
ro_out_max = 1.50004
ro_out_min = 0.000324079
ro_out_p2p = 1.49972
n0_p2p = 1.56761
n1_p2p = 1.55603
n2_p2p = 1.56051
n3_p2p = 1.50006
n4_p2p = 1.50056
n5_p2p = 1.43381
n6_p2p = 1.50084
n7_p2p = 1.49977
n8_p2p = 1.49999
n9_p2p = 1.43121
n10_p2p = 1.50052
```

```
vdda_lv = 1.5
--
idda_uA = 1015.61
freq_MHz = 8.91494
D_cyc = 51.7299
ro_out_max = 1.49836
ro_out_min = 0.0125407
ro_out_p2p = 1.48582
n0_p2p = 1.54397
n1_p2p = 1.54279
n2_p2p = 1.53435
n3_p2p = 1.48589
n4_p2p = 1.49982
n5_p2p = 1.44014
n6_p2p = 1.49981
n7_p2p = 1.47962
n8_p2p = 1.49952
n9_p2p = 1.43812
n10_p2p = 1.49927
```

Figure 19 fmin and fmax at mid code

Figure 19 shows f_{osc} = 8.92 → 12.35MHz, representing a spread of ± 16.2% which is inside the ± 19.5% tuning range of the CDAC.

Figure 20 demonstrates the tuning capability at the fmax corner where $f_{osc} = 12.36MHz$ (at trim code = 31) gets trimmed to $f_{osc} = 10.67MHz$ (at trim code = 63).

```
vdda_lv = 1.5
--
idda_uA = 346.109
freq_MHz = 12.355
D_cyc = 52.9189
ro_out_max = 1.50004
ro_out_min = 0.000324079
ro_out_p2p = 1.49972
n0_p2p = 1.56761
n1_p2p = 1.55603
n2_p2p = 1.56051
n3_p2p = 1.50006
n4_p2p = 1.50056
n5_p2p = 1.43381
n6_p2p = 1.50084
n7_p2p = 1.49977
n8_p2p = 1.49999
n9_p2p = 1.43121
n10_p2p = 1.50052
```

```
vdda_lv = 1.5
--
idda_uA = 370.724
freq_MHz = 10.6683
D_cyc = 52.3694
ro_out_max = 1.49994
ro_out_min = 0.022927
ro_out_p2p = 1.47701
n0_p2p = 1.56861
n1_p2p = 1.5608
n2_p2p = 1.56051
n3_p2p = 1.50228
n4_p2p = 1.50088
n5_p2p = 1.46804
n6_p2p = 1.50068
n7_p2p = 1.49985
n8_p2p = 1.50043
n9_p2p = 1.46693
n10_p2p = 1.49885
```

Figure 20 RO tuning at fmax corner

Figure 21 then demonstrates tuning capability at the fmin corner where $f_{osc} = 8.91MHz$ (at trim code = 31) gets trimmed to $f_{osc} = 11.58MHz$ (at trim code = 0).

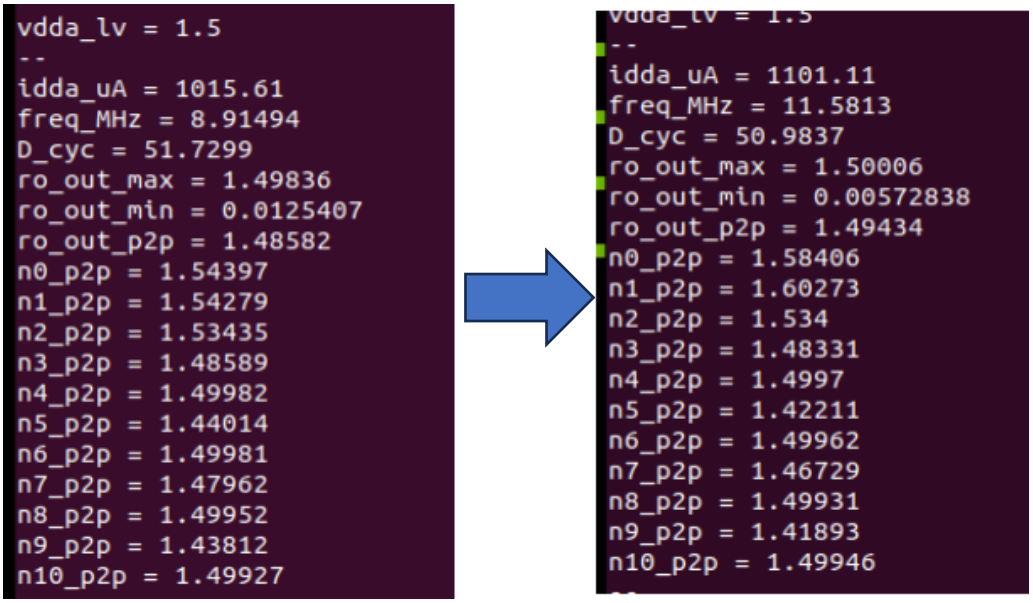

Figure 21 RO tuning at fmin corner

One will notice that the RO trims out the fmin corner better than the fmax corner. This is due to the following two reasons:

- As described in section 2, f_{max} occurs at R_{min}, C_{min} corner. Consequently LSBmin occurs at this corner, meaning more trim codes are needed to trim f_{max} back to its target frequency. The opposite is true for the f_{min} corner (occuring at R_{max}, C_{max}) where LSBmax occurs, meaning less trim codes are required.
- As detailed in section 5, the RO was re-centered at 10.56MHz (mid code at TT/27C). Due to the trim non-linearity, fmax = 16.9% larger (12.35MHz) while fmin = 15.7% smaller (8.91MHz). As a result, fmax is pushed 1.2% further from the target frequency than fmin.

One could argue the trim needs to be re-centered further because of this. However, as the fmax corner gets tuned to the target frequency pre layout, post layout parasitics will serve to push it closer making a pre layout re-centering unnecessary.

5.3 Frequency spread

Frequency spread is found by adding the spread in frequency across temperature ($\Delta f_{osc}(T)$) and supply variation ($\Delta f_{osc}(vdd)$) to the trim resolution, or LSB, as per (5).

$$\Delta f_{osc} = \Delta f_{osc}(T) + \Delta f_{osc}(vdd) + LSB_{trim} \dots (5)$$

$\Delta f_{osc}(T)$ from -40 → 125C is shown in Figure 22 where it can be found to give a spread of ± 2.73C, or a sensitivity of 3.4kHz / C.

$\Delta f_{osc}(vdd)$ for ± 5% supply variation is then shown in Figure 23 where it can be found to give a spread of ± 0.2C, or a sensitivity of 276Hz / mV.

As mentioned in section 2.1, inserting the RC network dramatically improves the PSR of the RO since without the RC elements, sensitivity would be 10 – 20MHz/mV i.e. 4 orders of magnitude worse.

Using these values gives:

$$\Delta f_{osc} = 2.73 + 0.2 + 0.67 = \pm 3.6\%$$

which is well within the ± 5% spec.

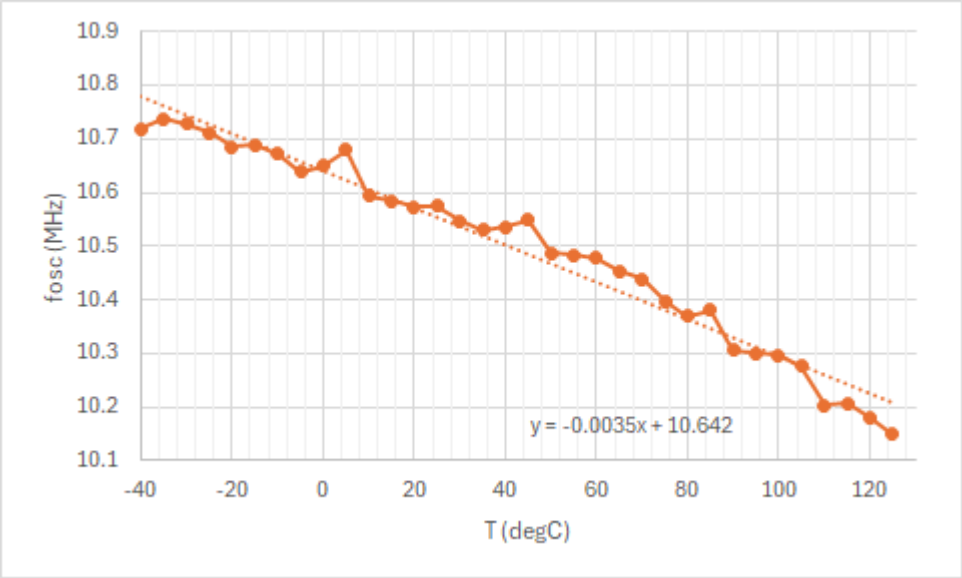

Figure 22 Frequency spread across temperature

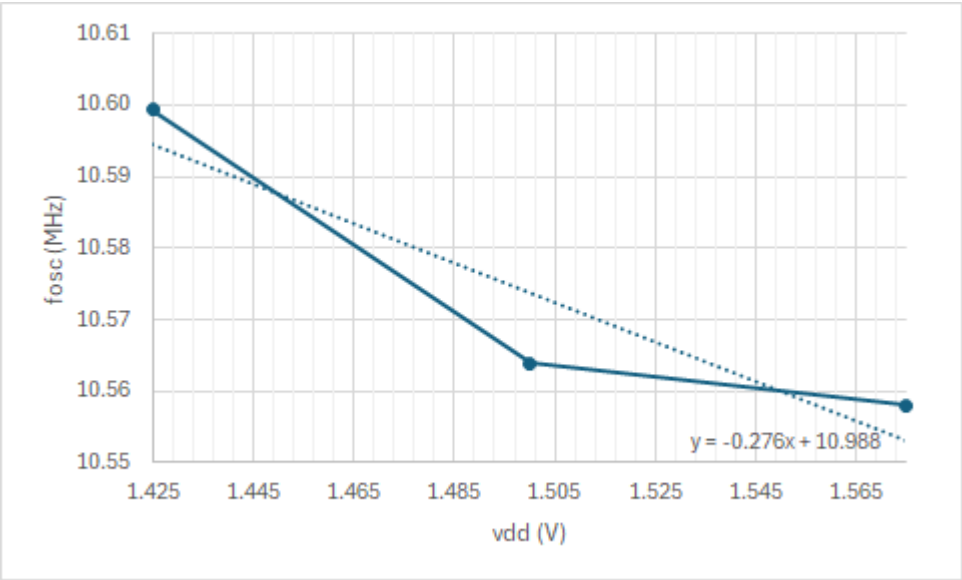

Figure 23 Frequency spread across supply variation

5.4 Duty cycle and power consumption

As Figure 20 and Figure 21 represent corner extremes, their reported power and duty cycle represents the full PVT spread of these parameters.

From Figure 20 we can see the max duty cycle = 52.9% (occurring at the fmax corner) which is within spec. The min duty cycle = 51% (occurring at the fmin corner) which is within spec.

From Figure 21 we can see the max current consumption = 1.1mA which is marginally outside the 1mA spec. As this is a loose spec however, there is no major concern here, especially since the value is expected to marginally improve post layout. In any case, once can stay within the specified power 1.5mW power spec by simply adjusting vdd to 1.36V at this corner (digital std cells can operate down to 1.2V with negligible change in f_{osc} due to this architectures superior PSR).

6 Conclusion

Table 3 summarises the performance of the block at DR2 w.r.t the specifications given in

Table 1. This shows the block to meeting all relevant specs thereby qualifying it for full schematic freeze and delivery to layout.

Design parameter	Unit	Spec	DR2	Comments
Fosc (typical conditions)	MHz	10	10.56	Intentionally made > 10MHz to allow for parasitics post layout
Fosc spread	± %	5	3.6	
Resolution	%	1	0.67	Average resolution
Dfosc	± %	20	19.5	Covers the ± 16.2% spread of the RO
Duty cycle max	%	55	52.9	
Duty cycle min	%	45	51	
idda_hv	mA	1	1.1	Marginal miss not a concern
Power	mW	1.5	1.65	vdd can be reduced to 1.36V to meet this with negligible change in fosc

Table 3: Design parameters required to meet target specifications

7 References

[1] https://ngspice.sourceforge.io/download.html

[2] https://xschem.sourceforge.io/stefan/xschem_man/install_xschem.html

[3] https://openvaf.semimod.de/

[4] https://github.com/IHP-GmbH/ihp-sg13cmos5l

[5] https://github.com/IHP-GmbH/IHP-Open-PDK

[6]https://www.ihp-microelectronics.com/services/research-and-prototyping-service/mpw-prototyping-service/schedule-price-list

8 Revision History

Rev.	Date	Author	Changes, Remarks
A	03.05.2026	Diarmuid Collins	Created

Table 4: Revision history